Adarsh Kumar T G

Software Engineer | C#, .NET, WPF, MVVM

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Portfolio: https://adarshkumar-tg.github.io/Adarsh-Kumar-T-G/

Professional Summary

Software Engineer with 2 years of experience developing Windows desktop applications using C#, .NET, WPF and MVVM. Work spans medical imaging and radiotherapy software, hardware integration, application communication, multi-monitor experiences, and TFS-based build and deployment automation.

Technical Skills

Languages: C#, SQL, JavaScript, HTML5, CSS3

Frameworks: .NET Framework, WPF, MVVM, Windows Services, Windows Forms

Architecture: OOP, IPC, Memory-mapped files, TCP/IP, Sockets, LINQ, Multi-monitor management

Database: SQL queries, Stored procedures, Large dataset handling

Medical Software: DICOM, TPS operations, Radiotherapy data handling, Medical image processing workflow

Hardware Integration: Fingerprint sensors, Touchscreens, Arduino, Wi-Fi communication

DevOps and Security: TFS 2018, CI/CD, Build agents, Automated testing, AES, SHA-512, DLL development, Setup packaging

Professional Experience

Software Engineer | Panacea Medical Technologies Pvt. Ltd.

2024 – Present | Bangalore, Karnataka, India

- Develop Windows desktop applications for medical imaging and radiotherapy using C#, .NET, WPF and MVVM, with 2 years of hands-on experience.

Projects

Mammography Software | C#, WPF, MVVM, IPC, TCP/IP

- Worked on UI, backend, workflows and data operations; connected processes via memory-mapped files and TCP/IP ports.
- Built multi-monitor support and a quarter-screen layout that shares the display with an image processing application.

Virtual Keyboard and Touchpad | C#, WPF

- Designed and built a virtual keyboard and a mouse-like touchpad from scratch for touchscreen use, integrated into the mammography application.

Fingerprint Sensor Integration | C#, TCP/IP, Sockets

- Independently implemented hardware-to-software communication using socket programming on a dedicated port.

Embedded System Web UI over Wi-Fi | HTML, CSS, JavaScript, Arduino

- Built a responsive web interface hosted on an embedded board to trigger and monitor Arduino-based hardware over Wi-Fi.

Krystal: Radiotherapy TPS Integration | C#, WPF, DICOM, SQL

- Exchanged patient, plan, image and dose data with treatment planning systems; generated dynamic PDF reports and live 3D visuals.

Common License Management System | C#, AES, SHA-512, DLL

- Built one reusable licensing solution shared across company projects, using AES encryption, SHA-512 verification and system-specific license generation.

TFS DevOps and CI/CD Automation | TFS, Pipelines, C#, WPF

- Automated build, setup generation, testing and production release; managed agents, pools and build definitions and troubleshot pipeline failures.

Education

Master of Computer Applications (MCA) | The Oxford College of Engineering, Bengaluru
June 2024 | CGPA: 7.01

Bachelor of Computer Applications (B.Sc. Computer Science) | The National College, Basavanagudi, Bengaluru
September 2022 | CGPA: 6.09

Pre-University (PCMCS, Computer Science) | BGS PU College, Chikkaballapur
March 2019 | CGPA: 8.00